

Tinotenda Musikavanhu

Frontend Engineer • AI Integration • Systems Design

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🌐 Portfolio • [LinkedIn](#) • [GitHub](#)

Professional Summary

Results-driven Frontend Engineer with 6+ years of experience building responsive, high-performance web applications. Specialising in **React**, **TypeScript**, and **Next.js**, with a strong track record delivering production systems under enterprise SLAs. Passionate about clean UX, AI-driven product features, and pushing the boundaries of what is possible on the web. Currently exploring latent space planning and autoregressive generation research via PyTorch and HuggingFace.

Education

Southern Methodist University

2016 – 2020

Bachelor of Science, Computer Science

Emphasis: Systems Engineering & Artificial Intelligence

Location: Dallas, TX

Harvard University — Extension School

Ongoing

Continuing Education

Coursework: Advanced Computational Theory, Ethical Frameworks in AI & Technology

Location: Cambridge, MA

Professional Experience

Application Engineer

2019 – 2025

ProtaTECH — Dallas, TX

- Deployed and maintained critical event applications for high-stakes operations including **Super Bowl** and large-scale public events, achieving **99.99% uptime** under real-time load.
- Engineered and optimised data pipelines and application interfaces for efficient real-time data capture and analysis in dynamic field environments.
- Provided on-site technical support and rapid incident response, managing complex networked systems under extreme operational pressure.
- Led frontend architecture decisions using **React** and **TypeScript** for internal tooling dashboards.

Applied AI Systems Developer

2021 – Present

Freelance & Independent Projects

- Built **GenesisTrader** — an AI-powered trading system integrating Coinbase APIs, predictive analytics, and real-time market intelligence with a Next.js frontend.
- Developed **Genesis** AI research assistant for autonomous workflow optimisation, multi-agent orchestration, and local LLM integration (LM Studio / DeepSeek).
- Created an **AI Gameplay Automation System** for Sims 4, using computer vision and reinforcement learning to autonomously control game agents — built in Python with a React monitoring dashboard.
- Deployed scalable AI safety testing pipelines and multi-agent systems using modern LLM tooling.

AI Systems Researcher

2023 – Present

Safehaven Labs — Open Science Initiative

- Conducting independent research into **Latent Planning + Autoregressive Generation**, training custom transformer architectures on Apple MPS hardware using PyTorch and HuggingFace.
- Exploring contrastive loss, attention map attribution, and diffusion model fine-tuning as mechanisms for improved latent space coherence.
- Authored research publications on Zenodo (500+ downloads) covering AI systems, climate modelling, and emergent computation.

Technical Skills

Frontend & Web

- React, Next.js, TypeScript
- Framer Motion, Tailwind CSS
- Responsive Design, Web Perf
- Vanilla CSS, HTML5, SVG

AI & Machine Learning

- PyTorch, HuggingFace Transformers
- Latent Space & Diffusion Models
- LLM Integration (LM Studio, OpenAI)
- Reinforcement Learning

Backend & Systems

- Node.js, REST APIs, WebSockets
- Distributed Systems Architecture
- Real-time Data Pipelines
- Cyber-Physical Resilience

Tools & Platforms

- Git, GitHub, Vercel, Netlify
- Apple MPS / GPU Training
- Figma, System Design
- Zenodo Open Science

Selected Projects

AI Gameplay Automation System (Sims 4)

Python, React, Computer Vision

Autonomous AI agent that plays and manages a Sims 4 household using computer vision, decision trees, and reinforcement learning. Features a real-time React monitoring dashboard with live game-state telemetry.

GenesisTrader

Next.js, TypeScript, Coinbase API

AI-powered cryptocurrency trading system with predictive analytics, real-time market intelligence, and a high-performance Next.js frontend. Integrates local LLM reasoning for trade signal generation.

AI Systems Research — Latent Planning + Autoregressive Gen

PyTorch, HuggingFace, MPS

Independent research training custom transformer architectures to explore latent planning as a mechanism for improved autoregressive text and image generation quality.

Digital Portfolio

Next.js, TypeScript, Framer Motion

High-performance editorial portfolio featuring sticky full-viewport art backdrops, glassmorphism cards, parallax animations, and an interactive AI research section.

Selected Publications

Full list available at: zenodo.org/Musikavanhu

- **Musikavanhu, T.** (2025). “SolarTherm-CarbCap (STCC): Integrated Solar-Thermal Carbon Capture for Climate Remediation.” DOI: [10.5281/zenodo.15711985](https://doi.org/10.5281/zenodo.15711985)
- **Musikavanhu, T.** (2025). “Void One: Emergent Quantum Computation in Room-Temperature Systems.” DOI: [10.5281/zenodo.15486760](https://doi.org/10.5281/zenodo.15486760)
- **Musikavanhu, T.** (2024). “Quantum Zeta Field Theory: Unifying Framework for Quantum Fields and Mathematical Structures.”

Awards & Recognition

- **Open Science Impact** (2024–2025): 500+ research downloads across 64+ publications on Zenodo
- **NASA Recognition** (2025): Acknowledged by former NASA Administrator Dan Goldin for research contributions
- **High-Stakes Ops** (2019–2025): Zero-downtime delivery for Super Bowl-scale event deployments at ProtaTECH

“Building the future — one component, one model, one breakthrough at a time.”